

國立中山大學 115 學年度 碩士班考試入學招生考試試題

科目名稱：計算機概論【資管系碩士班甲組、乙組】

—作答注意事項—

考試時間：100 分鐘

- 考試開始鈴響前不得翻閱試題，並不得書寫、劃記、作答。請先檢查答案卷（卡）之應考證號碼、桌角號碼、應試科目是否正確，如有不同立即請監試人員處理。
- 答案卷限用藍、黑色筆(含鉛筆)書寫、繪圖或標示，可攜帶橡皮擦、無色透明無文字墊板、尺規、修正液（帶）、手錶(未附計算器者)。每人每節限使用一份答案卷，請衡酌作答。
- 答案卡請以 2B 鉛筆劃記，不可使用修正液（帶）塗改，未使用 2B 鉛筆、劃記太輕或污損致光學閱讀機無法辨識答案者，後果由考生自負。
- 答案卷（卡）應保持清潔完整，不得折疊、破壞或塗改應考證號碼及條碼，亦不得書寫考生姓名、應考證號碼或與答案無關之任何文字或符號。
- 可否使用計算機請依試題資訊內標註為準，如「可以」使用，廠牌、功能不拘，唯不得攜帶書籍、紙張（應考證不得做計算紙書寫）、具有通訊、記憶、傳輸或收發等功能之相關電子產品或其他有礙試場安寧、考試公平之各類器材入場。
- 試題及答案卷（卡）請務必繳回，未繳回者該科成績以零分計算。
- 試題採雙面列印，考生應注意試題頁數確實作答。
- 違規者依本校招生考試試場規則及違規處理辦法處理。

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※本科目依簡章規定「不可以」使用計算機(選擇題)

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很fresh笑死，僅供參考

單選題，每題 5 分。

1. Which of the following best describes the characteristics of packet switching?
A. A dedicated connection must be established before data transmission circuit
C B. All packets follow the same transmission path
C C. Each packet can be routed independently
D D. No routers are required in the network
2. Which of the following protocols is NOT an application-layer protocol?
A. HTTP
B B. ARP
C C. SMTP
D D. DNS
3. During the TCP three-way handshake, which of the following statements is correct?
A. The second packet has both the SYN and ACK flags set
A B. The client acknowledges the server's sequence number in the first packet
C C. The primary purpose of the three-way handshake is flow control
D D. UDP uses the same handshake procedure
4. Which of the following is NOT a necessary condition for a deadlock to occur?
A. Mutual exclusion
B. Hold and wait
C C. With preemption 不可搶奪，意旨系統不可臨時重新配置資源
D D. Circular wait
5. In a virtual memory system, when a page fault occurs, which of the following actions is NOT necessarily performed?
A. Loading the page into main memory
B. Updating the page table
D C. Triggering an operating system interrupt
D D. Performing page replacement
6. Which of the following best describes pass-by-reference?
A. Passing the value of a variable
B B. Passing the address of a variable
C C. Passing the data type of a variable
D D. Passing a copy of a variable
7. Which of the following systems is most suitable for using a NoSQL database rather than a relational database?
A. Core banking transaction systems
B B. Social media user activity data
C C. Payroll management systems
D D. Accounting general ledger systems
8. Which of the following scenarios is most suitable for using multithreading rather than multiprocessing?
A A. Tasks that frequently share data and require fast context switching

試題請隨卷繳回，請留意背面是否有題

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- B. Tasks that require strong isolation and high security
C. Tasks that require parallel computation across multiple machines
D. Systems that must avoid any synchronization mechanisms
9. Which of the following statements most accurately describes Big-O notation in time complexity analysis?
A. Big-O describes the average-case running time for all inputs
D B. Big-O provides the exact upper bound of actual execution time
C. Big-O applies only to recursive algorithms
D. Big-O ignores constants and lower-order terms to describe the asymptotic upper bound
10. In object-oriented programming, which of the following statements about polymorphism is most accurate?
A. Polymorphism occurs only at compile time and is unrelated to runtime behavior
B B. Polymorphism allows the actual method to be determined at runtime
C. Polymorphism is equivalent to method overloading
D. The purpose of polymorphism is to improve program execution performance

Question Group 1 (15%, each 5%)

Consider the following C program fragment, and answer question 11 to 13. Some parts are intentionally omitted.

```
int* process(int n) {
    int *arr = (int*) malloc(n * sizeof(int));
    if (arr == NULL) return NULL;

    for (int i = 0; i < n; i++) {
        arr[i] = i * 2;
    }

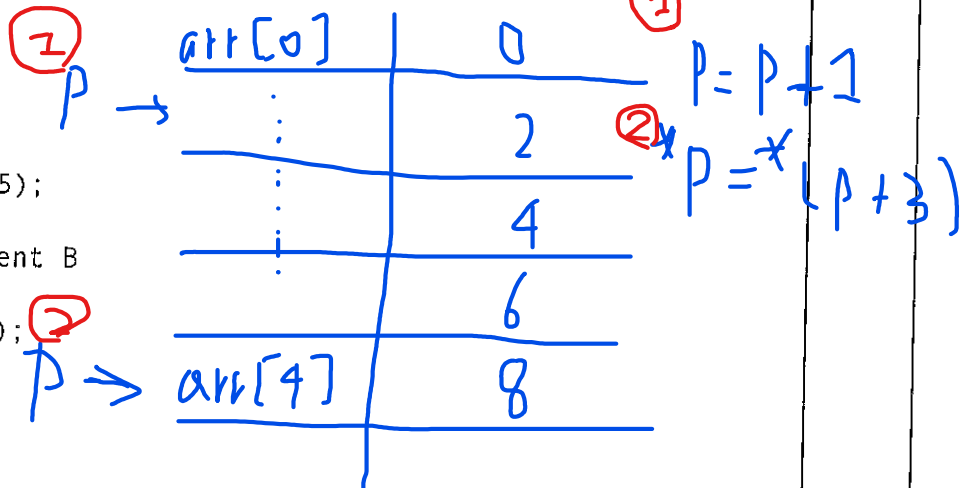
    // Missing statement A: ensure arr can store one extra int

    arr[n] = -1;
    return arr;
}

int main() {
    int *p = process(5);

    // Missing statement B

    printf("%d\n", *p);
    free(p);
    return 0;
}
```



11. Which option is the best completion for **statement A** to ensure correctness and avoid unnecessary overhead?

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A.

```
int *tmp = (int*)realloc(arr, (n + 1) * sizeof(int));  
if (tmp == NULL) { free(arr); return NULL; }  
arr = tmp;
```

B.

```
arr = (int*)realloc(arr, (n + 1) * sizeof(int));
```

A

C.

```
int *tmp = (int*)malloc((n + 1) * sizeof(int));  
for (int i = 0; i < n; i++) tmp[i] = arr[i];  
arr = tmp;
```

D.

```
arr = (int*)realloc(arr, (n + 1) * sizeof(int*));
```

12. Assume the programmer chose the best answer for statement A. Which memory-related issue can still occur in the full program? **heap可能不夠做動態分配**

C

- A. The program has a use-after-free because `free(p)` is executed before `printf`
- B. The program will always double-free because `process()` frees `arr` and `main()` frees it again
- C. If memory allocation fails, `main()` may dereference a NULL pointer in `printf("%d\n", *p);`
- D. The statement `arr[n] = -1;` always causes a buffer overflow because `arr` has length `n`

13. Suppose Missing **statement B** is completed as below, what value will be printed?

```
p = p + 1;  
*p = *(p + 3);
```

C

- A. 0
- B. 2
- C. 8
- D. -1

Question Group 2 (20%, each 5%)

Consider the following diagram shows an Agentic Large Language Model (LLM) System, and answer question 14 to 17.

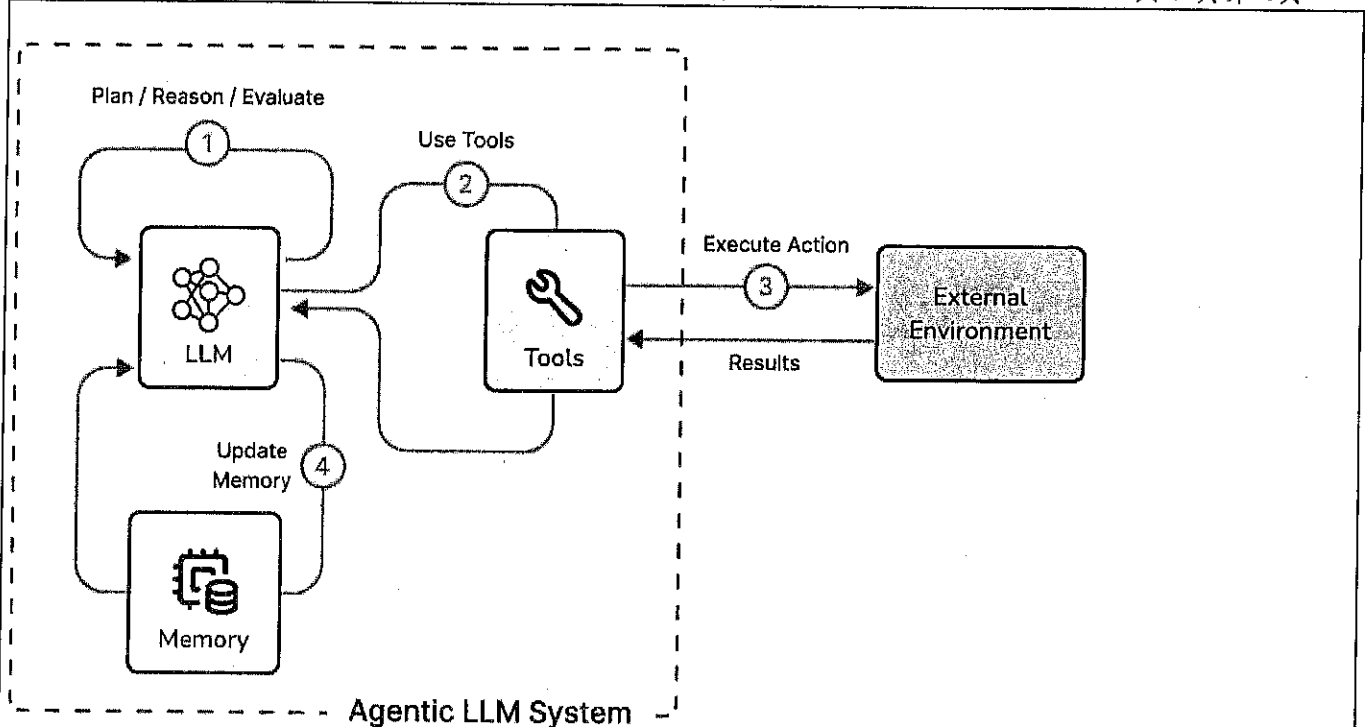
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14. Which neural network architecture is most commonly used for text generation in modern LLMs?
- A. Generative Adversarial Network
 - B. Long Short-Term Memory Recurrent Neural Network
 - C. Variational Autoencoder
 - D. Decoder-only Transformer
15. A user asks an agentic LLM: "Find the current stock price of TSMC and calculate how much 100 shares would cost in USD and TWD." Assume the system has access to a stock API and a currency conversion API. Which behavior correctly follows the agentic workflow?
- A. Call stock API → Call currency API → Calculate both values in parallel → Return results
 - B. Identify two sub-tasks → Call stock API → Evaluate result → Plan currency conversion → Call currency API → Calculate → Verify → Return results
 - C. Call stock API and currency API simultaneously → Combine results → Return answer
 - D. Retrieve stock price and exchange rate from memory → Calculate → Verify with tools → Return results
16. Which of the following techniques is generally NOT intended to reduce hallucination in an agentic AI system?
- A. Increasing sampling temperature
 - B. Retrieval-Augmented Generation (RAG)
 - C. Tool-based verification
 - D. Self-reflection or consistency checking

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17. Which of the following is NOT commonly encountered as an attack in an agentic LLM system?

- A. Prompt injection
- B. Model inversion
- C. Membership inference
- D. Least-privilege tool access control

Question Group 3 (15%, each 5%)

Consider the output of a student's `ipconfig` command below, and answer question 18 to 20.

```
IPv4 Address . . . . . : 192.168.1.23
Subnet Mask . . . . . : [hidden]
Default Gateway . . . . . : 192.168.1.1
```

18. Which operating system is **most likely** being used?

- A. FreeBSD
- B. Windows (PowerShell/Command Prompt)
- C. Ubuntu
- D. macOS

19. Assuming a typical home or campus LAN configuration, what is the most likely subnet mask?

- A. 255.0.0.0
- B. 255.255.0.0
- C. 255.255.255.0
- D. 255.255.255.255

20. Which device is responsible for assigning the IP address shown above?

- A. Internet Service Provider (ISP)
- B. DNS Server
- C. DHCP Server
- D. Network Interface Card (NIC)

動態ip地址分配，分配的是私有ip